

DBMS LAB 4 : BANK DATABASE

1. DB AND TABLE CREATION

```
1 • CREATE DATABASE bank;
2 • USE bank;
3
4 • CREATE TABLE Branch (
5     Branchname VARCHAR(50) PRIMARY KEY,
6     Branchcity VARCHAR(50) NOT NULL,
7     Assets DECIMAL(15,2) NOT NULL
8 );
9
10 • CREATE TABLE BankCustomer (
11     Customername VARCHAR(50) PRIMARY KEY,
12     Customerstreet VARCHAR(100) NOT NULL,
13     City VARCHAR(50) NOT NULL
14 );
15
16 • CREATE TABLE BankAccount (
17     Accno INT PRIMARY KEY,
18     Branchname VARCHAR(50) NOT NULL,
19     Balance DECIMAL(15,2) NOT NULL,
20     FOREIGN KEY (Branchname) REFERENCES Branch(Branchname)
21 );
22
```

```
23 ● ○ CREATE TABLE Depositer (  
24     Customername VARCHAR(50),  
25     Accno INT,  
26     PRIMARY KEY (Customername, Accno),  
27     FOREIGN KEY (Customername) REFERENCES BankCustomer(Customername),  
28     FOREIGN KEY (Accno) REFERENCES BankAccount(Accno)  
29 );  
30  
31 ● ○ CREATE TABLE Loan (  
32     Loannumber INT PRIMARY KEY,  
33     Branchname VARCHAR(50) NOT NULL,  
34     Amount DECIMAL(15,2) NOT NULL,  
35     FOREIGN KEY (Branchname) REFERENCES Branch(Branchname)  
36 );
```

2. INSERTING VALUES

```
37
38 • INSERT INTO Branch (Branchname, Branchcity, Assets) VALUES
39 ('SBI_Chamrajpet', 'Bangalore', 50000),
40 ('SBI_ResidencyRoad', 'Bangalore', 10000),
41 ('SBI_ShivajiRoad', 'Bombay', 20000),
42 ('SBI_ParlimentRoad', 'Delhi', 10000),
43 ('SBI_Jantarmanatar', 'Delhi', 20000);
44
45 • INSERT INTO BankCustomer (Customername, Customerstreet, City) VALUES
46 ('Avinash', 'Bull_Temple_Road', 'Bangalore'),
47 ('Dinesh', 'Bannerghatta_Road', 'Bangalore'),
48 ('Mohan', 'NationalCollege_Road', 'Bangalore'),
49 ('Nikil', 'Akbar_Road', 'Delhi'),
50 ('Ravi', 'Prithviraj_Road', 'Delhi');
51
52 • INSERT INTO BankAccount (Accno, Branchname, Balance) VALUES
53 (1, 'SBI_Chamrajpet', 2000),
54 (2, 'SBI_ResidencyRoad', 5000),
55 (3, 'SBI_ShivajiRoad', 6000),
56 (4, 'SBI_ParlimentRoad', 9000),
57 (5, 'SBI_Jantarmanatar', 8000),
58 (6, 'SBI_ShivajiRoad', 4000),
59 (8, 'SBI_ResidencyRoad', 4000),
```

```
64 ● INSERT INTO Depositer (Customername, Accno) VALUES
65     ('Avinash', 1),
66     ('Dinesh', 2),
67     ('Nikil', 4),
68     ('Ravi', 5),
69     ('Avinash', 8),
70     ('Nikil', 9),
71     ('Dinesh', 10),
72     ('Nikil', 11);
73
74 ● INSERT INTO Loan (Loannumber, Branchname, Amount) VALUES
75     (1, 'SBI_Chamrajpet', 1000),
76     (2, 'SBI_ResidencyRoad', 2000),
77     (3, 'SBI_ShivajiRoad', 3000),
78     (4, 'SBI_ParlimentRoad', 4000),
79     (5, 'SBI_Jantarmantar', 5000);
80
```

3. VIEWING ALL TABLES

A. BRANCH

```
82 • SELECT * FROM Branch;
83
```

Branchname	Branchcity	Assets
SBI_Chamrajpet	Bangalore	50000.00
SBI_Jantarantar	Delhi	20000.00
SBI_ParliamentRoad	Delhi	10000.00
SBI_ResidencyRoad	Bangalore	10000.00
SBI_ShivajiRoad	Bombay	20000.00
NULL	NULL	NULL

B. BANK CUSTOMER

```
85 • SELECT * FROM BankCustomer;
86
```

Customername	Customerstreet	City
Avinash	Bull_Temple_Road	Bangalore
Dinesh	Bannergatta_Road	Bangalore
Mohan	NationalCollege_Road	Bangalore
Nikil	Akbar_Road	Delhi
Ravi	Prithviraj_Road	Delhi
NULL	NULL	NULL

C. BANK ACCOUNT

```
88 • SELECT * FROM BankAccount;
89
```

Accno	Branchname	Balance
1	SBI_Chamrajpet	2000.00
2	SBI_ResidencyRoad	5000.00
3	SBI_ShivajiRoad	6000.00
4	SBI_ParliamentRoad	9000.00
5	SBI_Jantarantar	8000.00
6	SBI_ShivajiRoad	4000.00
8	SBI_ResidencyRoad	4000.00
9	SBI_ParliamentRoad	3000.00
10	SBI_ResidencyRoad	5000.00
11	SBI_Jantarantar	2000.00

D. DEPOSITER

```
91 • SELECT * FROM Depositer;
92
```

Customername	Accno
Avinash	1
Dinesh	2
Nikil	4
Ravi	5
Avinash	8
Nikil	9
Dinesh	10
Nikil	11
NULL	NULL

E. LOAN

94 ● `SELECT * FROM Loan;`

95

Result Grid



Filter Rows:

Edi

	Loannumber	Branchname	Amount
▶	1	SBI_Chamrajpet	1000.00
	2	SBI_ResidencyRoad	2000.00
	3	SBI_ShivajiRoad	3000.00
	4	SBI_ParlimentRoad	4000.00
	5	SBI_Jantarmantar	5000.00
★	NULL	NULL	NULL

4. EXECUTING QUERIES

- A. Display the branch name and assets from all branches in lakhs of rupees and rename the assets column to 'assets in lakhs'.

```
96 • SELECT Branchname, Assets/100000 AS 'Assets in lakhs'
97 FROM Branch;
98
```

Result Grid   Filter Rows: Export:  Wrap Cel

	Branchname	Assets in lakhs
▶	SBI_Chamrajpet	0.500000
	SBI_Jantarmantar	0.200000
	SBI_ParlimentRoad	0.100000
	SBI_ResidencyRoad	0.100000
	SBI_ShivajiRoad	0.200000

B. Find all the customers who have at least two accounts at the same branch (ex.SBI_ResidencyRoad).

```
99 • SELECT d.Custormername, ba.Branchname, COUNT(*) AS AccountCount
100 FROM Depositer d
101 JOIN BankAccount ba ON d.Accno = ba.Accno
102 GROUP BY d.Custormername, ba.Branchname
103 HAVING COUNT(*) >= 2;
104
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	Customername	Branchname	AccountCount		
▶	Dinesh	SBI_ResidencyRoad	2		
	Nikil	SBI_ParlimentRoad	2		

c. CREATE A VIEW WHICH GIVES EACH BRANCH THE SUM OF THE AMOUNT OF ALL THE LOANS AT THE BRANCH.

```
105 • CREATE VIEW Branch_Loan_Summary AS
106 SELECT Branchname, SUM(Amount) AS Total_Loan_Amount
107 FROM Loan
108 GROUP BY Branchname;
109 • SELECT * FROM Branch_Loan_Summary;
```

Result Grid		Filter Rows:	Export:	Wrap C
	Branchname	Total_Loan_Amount		
▶	SBI_Chamrajpet	1000.00		
	SBI_Jantarmanatar	5000.00		
	SBI_ParliamentRoad	4000.00		
	SBI_ResidencyRoad	2000.00		
	SBI_ShivajiRoad	3000.00		

ADDITIONAL QUERIES

1. Find all the customers who have an account at all the branches located in a specific city (Ex. Delhi).

```
112 • SELECT DISTINCT d.Customername
113 FROM Depositer d
114 JOIN BankAccount ba ON d.Accno = ba.Accno
115 JOIN Branch b ON ba.Branchname = b.Branchname
116 WHERE b.Branchcity = 'Delhi'
117 GROUP BY d.Customername
118 HAVING COUNT(DISTINCT ba.Branchname) = (
119     SELECT COUNT(*) FROM Branch WHERE Branchcity = 'Delhi'
120 );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	Customername
▶	Nikil

2. Find all customers who have a loan at the bank but do not have an account.

```
122 • SELECT DISTINCT bc.Customername
123 FROM BankCustomer bc
124 WHERE bc.Customername IN (
125     SELECT DISTINCT Customername FROM Depositer
126 )
127 AND bc.Customername NOT IN (
128     SELECT DISTINCT Customername FROM Depositer
129 );
```

Result Grid

	Customername
*	HULL

3. Find all customers who have both an account and a loan at the Bangalore branch

```
131 SELECT DISTINCT d.Custormername
132 FROM Depositer d
133 JOIN BankAccount ba ON d.Accno = ba.Accno
134 JOIN Branch b ON ba.Branchname = b.Branchname
135 WHERE b.Branchcity = 'Bangalore'
136 AND d.Custormername IN (
137     SELECT bc.Custormername
138     FROM BankCustomer bc
139     WHERE EXISTS (
140         SELECT 1 FROM Loan l
141         JOIN Branch b2 ON l.Branchname = b2.Branchname
142         WHERE b2.Branchcity = 'Bangalore'
143     )
144 );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

	Custormername
▶	Avinash
	Dinesh

4, Find the names of all branches that have greater assets than all branches located in Bangalore.

```
146 • SELECT Branchname
147 FROM Branch
148 WHERE Assets > ALL (
149     SELECT Assets FROM Branch WHERE Branchcity = 'Bangalore'
150 );
151
```

Result Grid | Filter Rows: | Edit: | Export/

Branchname
1000

5. Demonstrate how you delete all account tuples at every branch located in a specific city (Ex. Bombay).

```
152 • DELETE FROM Depositer
153 WHERE Accno IN (
154     SELECT Accno FROM BankAccount ba
155     JOIN Branch b ON ba.Branchname = b.Branchname
156     WHERE b.Branchcity = 'Bombay'
157 );
158
```

6. Update the Balance of all accounts by 5%

```
164 • UPDATE BankAccount  
165   SET Balance = Balance * 1.05;  
166 • select * from BankAccount
```

Result Grid			
Filter Rows:			
Acno	Branchname	Balance	
1	SBI_Chamrajpet	2100.00	
2	SBI_ResidencyRoad	5250.00	
3	SBI_ShivajRoad	6300.00	
4	SBI_ParliamentRoad	9450.00	
5	SBI_Jantarmantar	8400.00	
6	SBI_ShivajRoad	4200.00	
8	SBI_ResidencyRoad	4200.00	
9	SBI_ParliamentRoad	3150.00	
10	SBI_ResidencyRoad	5250.00	
11	SBI_Jantarmantar	2100.00	
* NULL	NULL	NULL	